DEPT OF COMPUTER SCIENCE AND ENGINEERING
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This is to certify that the project entitled is hereby **"AI-Powered Web & File Security Shield" Tagline: A browser-based defense system detecting phishing links and malicious downloads.** being submitted by

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In partial fulfillment of the requirement for the award of the **Bachelor of Technology** in **Computer Science and Engineering** from **JNTUA College of Engineering , Pulivendula** during the academic year 2025-2026.

Brief Report of the Final Project Work Carried:

Abstract-

Artificial Intelligence (AI) plays a vital role in enhancing cybersecurity by enabling systems to detect, analyze, and prevent online threats in real time. In this project, AI is utilized to identify phishing websites, malicious links, and harmful file downloads that pose serious risks to users while browsing the internet. Machine Learning algorithms analyze URL structures, website behavior, file metadata, and historical threat patterns to classify content as safe or malicious.

The proposed project is an advanced browser-based security shield that continuously monitors web activities and file downloads. Whenever a user accesses a suspicious website or attempts to download a potentially harmful file, the system instantly analyzes the content and alerts the user before any damage occurs. Unlike traditional security tools that rely only on static blacklists, this system uses intelligent pattern recognition to detect new and unknown threats. A centralized web dashboard is developed using full-stack technologies where users can view detected threats, manage security settings, and review browsing safety history.

Furthermore, the project includes a file security module that scans downloaded files for malware signatures and abnormal behavior before allowing access. The combination of AI-based threat detection, browser-level monitoring, and intelligent file security makes this project a comprehensive, proactive, and real-world applicable solution for modern web security challenges.

Signature of Guide

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